

Lingshu - Project Progress Report

Generated: 2026-07-05

1. Completed Features

Phase 1: Core Infrastructure

- MCP Tool Registration CodeAnalysisTool in ToolRegistry
- Graph REST API GET/POST /v1/graph/{project}
- Project List API GET /v1/projects
- Visualization WebUI /v1/graph/{project}/view

Phase 2: Real-time Capabilities

- WebSocket Push graph.updated event on analysis
- SSE Push Real-time updates to WebUI
- SSE Error Handling Graceful warn! when no subscribers

Phase 3: Persistence & Watching

- NotifyFileObserver notify crate v6, native FS events
- PollingFileObserver Cross-platform polling fallback
- Auto-fallback Notify -> Polling on failure
- SQLite GraphStore Persist to ~/.local/share/lingshu/graphs.db
- Startup Recovery Auto-restore graphs from SQLite on boot

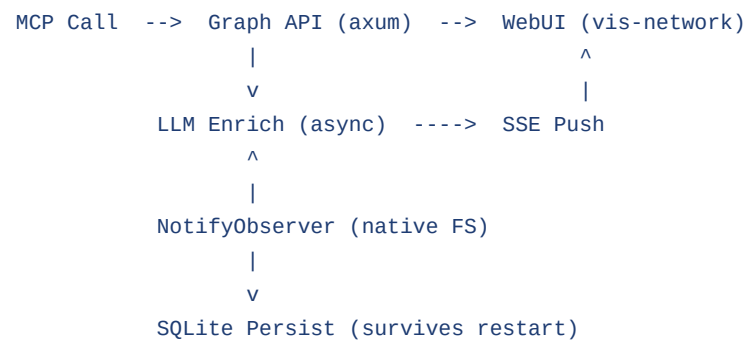
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2. Performance Benchmark

Files scanned:	2
Graph nodes:	90
Graph edges:	88
Pipeline duration:	~210ms
Node types:	Function, Class, File
DB size:	~4KB

3. Architecture



4. Files Changed

New files:

- knowledge-graph/src/store.rs -- GraphStore (SQLite)

Modified files:

- app/src/api.rs -- Graph routes, persistence save
- app/src/main.rs -- graph_cache, graph_store in runtime
- code-analyzer/src/observer.rs -- NotifyFileObserver
- code-analyzer/src/lib.rs -- New type exports
- orchestrator/src/pipeline.rs -- Integrated NotifyFileObserver
- websocket/src/broadcast.rs -- SSE log level fix
- knowledge-graph/Cargo.toml -- rusqlite dependency
- knowledge-graph/src/lib.rs -- GraphStore export

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5. Test Results

```
cargo test -p lingshu-knowledge-graph passed
cargo test -p lingshu-code-analyzer29 passed
cargo test -p lingshu-orchestrator24 passed
```

6. Next Steps

- B: MCP experience -- streaming, status query tool
- D: Error handling -- LLM fallback, watcher recovery
- Full scan optimization -- lift 5-file limit for large projects
- Docker deployment -- one-click container startup